Clinical Imaging 139 (2026) 110923

Contents lists available at ScienceDirect

Clinical Imaging

journal homepage: www.elsevier.com/locate/clinimag

Breast Imaging

Artificial intelligence–supported double reading in European population breast cancer screening: A systematic review and meta-analysis of prospective programs

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ARTICLE INFO

Keywords:
Artificial intelligence
Breast cancer screening
Mammography
Digital breast tomosynthesis
Double reading
Cancer detection rate
Recall rate
Population screening
Meta-analysis
Screening programs

ABSTRACT

Background: Most European population mammography screening programs rely on double reading with arbitration, a model that delivers mortality benefit but is increasingly challenged by radiologist workload, variable specificity, and interval cancers. Artificial intelligence (AI) is being evaluated to support or optimize these established European screening pathways.

Purpose: To synthesize prospective or program-embedded evaluations of AI conducted within European-style population screening programs and to estimate exploratory program-level absolute risk differences (RDs) per 1000 examinations for cancer detection rate (CDR) and recall.

Materials and methods: We performed a prespecified, focused evidence synthesis of three large studies embedded within routine population screening programs operating under European-relevant workflows: MASAI (randomized AI-supported risk triage within a national program), ScreenTrustCAD (prospective paired-reader evaluation with AI as an independent reader in a double-reading framework), and PRAIM (nationwide decision-referral implementation). Outcomes were harmonized as AI–control RDs per 1000 examinations. Random-effects pooling used Hartung–Knapp–Sidik–Jonkman models. For the paired-reader design, sensitivity analyses applied a Kish effective sample-size approach across plausible within-examination correlations (ρ = 0.3–0.8). Positive predictive value (PPV) and workflow/time outcomes were summarized descriptively.

Results: Across 597,419 examinations, the pooled CDR RD was +0.9 per 1000 (95% CI −0.0 to +1.8; $I^2 \approx 12\%$), consistent with a modest directional increase with borderline statistical uncertainty. The pooled recall RD was −0.6 per 1000 (95% CI −3.1 to +2.1; $I^2 \approx 41$–43%), indicating no consistent recall increase across screening programs. Where reported, PPV was higher with AI-supported screening. Efficiency signals included 44.3% fewer total readings in MASAI and shorter reading times for AI-normal examinations in PRAIM; in PRAIM, a program-level safety-net mechanism recovered 204 cancers that would otherwise have been missed.

Conclusion: In European population screening programs characterized by double reading and arbitration, prospective program-embedded evidence suggests that AI integration may yield a small absolute increase in cancer detection (≈1/1000) without a consistent increase in recall, alongside improved PPV and efficiency signals. These findings suggestAI primarily as a complementary reader within European screening workflows, with implementation requiring explicit quality assurance and monitoring of interval cancers and stage distribution.

1. Introduction

Population mammography screening reduces breast cancer mortality but remains constrained by increasing radiologist workload, variable specificity, and the persistence of interval cancers. In most European programs, screening is organized around double reading with arbitration, a resource-intensive model that delivers high cancer detection but is increasingly difficult to sustain at scale. This model requires substantial radiologist workload and may contribute to false-positive recalls; workforce constraints further strain its sustainability. Interval

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https://doi.org/10.1016/j.clinimag.2026.110923
Received 28 March 2026; Received in revised form 20 July 2026; Accepted 10 August 2026
Available online 11 August 2026
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cancers (often more prognostically unfavorable than screen-detected cancers) persist despite double reading, and ~20–30% show suspicious findings in hindsight on the prior screening mammogram.[1–9]

Artificial intelligence (AI) has been proposed as a means to support established screening workflows rather than replace radiologists outright. Early evaluations of AI in breast screening have largely focused on standalone diagnostic accuracy using retrospective or enriched datasets, limiting their relevance to population-based programs. More recently, prospective and program-embedded studies have examined AI integrated into routine screening pathways, including AI as an independent second reader, AI-based triage to reduce second human reads, and decision-referral models with safety-net mechanisms.[10–19] However, the absolute program-level effects of these strategies on cancer detection and recall remain incompletely quantified.

We therefore performed a focused evidence synthesis of prospective or prospectively embedded evaluations of AI conducted within European-relevant population screening programs. By harmonizing outcomes as absolute risk differences per 1000 examinations, we aimed to estimate the near-term tradeoffs associated with integrating AI into double-reading workflows, with particular attention to cancer detection, recall, and efficiency. We hypothesized that AI-supported workflows would increase cancer detection without a consistent recall penalty while improving workflow efficiency. Because these studies evaluate different human-AI implementation strategies, the pooled estimates should be interpreted as exploratory program-level benchmarks rather than as a single algorithm-specific treatment effect.

2. Materials and methods

2.1. Study design and oversight

We performed a focused evidence synthesis of prospective or prospectively embedded evaluations of AI implemented within European organized population screening programs or European-style double-reading workflows. Sources included PubMed/MEDLINE and ClinicalTrials.gov, screened from database inception to June 2025 using terms for ('mammograph*' OR 'breast screening') AND ('artificial intelligence' OR 'machine learning' OR specific product names) AND (trial OR prospective OR implementation). We also screened reference lists of key reviews and eligible studies.

Records were screened in duplicate at title/abstract and full text against prespecified criteria (population screening, concurrent standard-of-care comparator, prospective/real-world data collection, reportable CDR and recall per 1000).

2.2. Data sources and eligibility criteria

We included contemporary prospective or prospectively embedded evaluations that met all of the following criteria: (1) asymptomatic, population-based mammography screening; (2) AI-supported screening embedded in routine workflow with a concurrent standard-of-care comparator; (3) prospective or real-world data collection; (4) reporting of cancer detection rate and recall rate, or sufficient data to derive them; (5) large-scale evaluation, defined as at least 50,000 screening examinations; and (6) implementation within European organized population screening programs or European-style double-reading workflows.

The threshold of 50,000 examinations was selected because absolute differences in cancer detection and recall per 1000 examinations require large denominators to provide stable, program-level estimates. Smaller retrospective, enriched, reader-study, or simulation-only evaluations were excluded because they do not provide directly comparable estimates of cancer detection and recall in routine screening populations.

The restriction to European programs or European-style workflows was based on workflow comparability rather than geography alone. The review specifically addresses AI integration into organized screening systems characterized by double reading, arbitration or consensus, and program-level quality assurance.

Handling of multiple reports from the same study. When multiple publications described the same registered trial or overlapping participant cohorts, the reports were linked under a single study identifier and were not treated as independent studies. Only one report contributed quantitative data for each primary outcome, and overlapping participants were not included more than once. The report supplying the study-level data used in the original quantitative synthesis was retained as the primary data source, whereas later overlapping publications were treated as companion reports and used to supplement study methods, expanded-cohort findings, tumor characteristics, or follow-up outcomes. For MASAI, the 2023 prespecified clinical-safety analysis supplied the cancer-detection and recall data used in the primary meta-analysis; the 2025 protocol-defined expanded-cohort report was summarized narratively as companion evidence.

2.3. Outcomes and definitions

The primary quantitative outcomes were:

- Cancer detection rate (CDR): number of screen-detected cancers per 1000 women screened.
- Recall rate: number of recalls per 1000 women screened.

Secondary outcomes summarized narratively included PPV of recall/biopsy and workload (number of screen readings/consensus and reading time).

2.4. Data extraction and harmonization

Two reviewers abstracted study characteristics, AI workflow, denominators, and endpoints from the main tables and flow diagrams; an independent analyst verified all denominators and totals against the published tables.

To enable pooling on a public-health scale, we harmonized all endpoints to risk differences per 1000 (AI – control). When needed, we converted the trial's counts to per-1000 scale using the study denominator and the AI + one vs two radiologists' contrast, which most closely mirrors the other studies' AI-supported double reading vs standard comparisons.

2.5. Statistical analysis

Analyses used R (v4.3.x) and the meta package (v8.2-1), with Hartung–Knapp–Sidik–Jonkman random-effects as primary and a common-effect model for context. A fixed random seed was used to reproduce interval estimates. The de-identified extraction sheet and code that generate all study-level effects, pooled estimates, I^2, and figures are provided in the Supplement. Heterogeneity was summarized by I^2 and τ^2.[24]

2.6. Unit-of-analysis correction for the paired design

For the paired-reader study with within-examination correlation, we adjusted its information content using a Kish design effect: $Deff = 1 + (m - 1)\rho$ with $m = 2$ readers; the effective sample size is $N_eff = N / Deff$. This preserves the per-1000 RD point estimate while inflating its SE to reflect correlation. We examined $\rho \in \{0.3, 0.5, 0.8\}$, giving $Deff = \{1.3, 1.5, 1.8\}$ and N_eff for ScreenTrustCAD ($N = 55{,}581$) of 42,755, 37,054, and 30,879, respectively from published counts (MASAI, ScreenTrustCAD) or model-based rates (PRAIM). For ScreenTrustCAD, the RD SEs were inflated by the N_eff factor above; for PRAIM, we used the reported model-based point estimates and CIs to obtain SEs on the per-1000 scale.[20–22,24]

2.7. Sensitivity analyses

We (i) varied ρ over 0.3–0.8 for ScreenTrustCAD; (ii) performed leave-one-out analyses excluding ScreenTrustCAD or PRAIM; and (iii) reported both random- and common-effect models. All numerical outputs (individual-study RDs/SEs, pooled estimates, I^2, and leave-one-out results) are compiled in the Supplementary Materials.

We derived standard errors for PRAIM's model-based per-1000 rates from reported 95% CIs using SE ≈ (upper − lower) / (2 × 1.96), noting these reflect model specification uncertainty. For MASAI and ScreenTrustCAD, we computed RDs and SEs from observed counts.

Because the Kish effective sample-size correction reduces the information contribution of ScreenTrustCAD, we also performed a sensitivity analysis using the nominal ScreenTrustCAD sample size without Kish adjustment. Because ScreenTrustCAD used a paired-reader design in which the same examinations contributed to more than one reader-strategy comparison, the primary analysis adjusted its information contribution using a Kish effective sample-size approach. This preserved the published per-1000 risk-difference point estimates while inflating standard errors across plausible within-examination correlations. We evaluated ρ values of 0.3, 0.5, and 0.8. In response to uncertainty regarding this adjustment, we also performed a sensitivity analysis using the nominal ScreenTrustCAD sample size without Kish adjustment. Leave-one-out analyses were performed to evaluate the influence of each included study.

2.8. Ethical approval

This synthesis used published aggregate data only; no new human subjects' research was conducted. Each included study obtained local ethics approvals with consent waivers where applicable.

2.9. Software and reproducibility

Meta-analytic computations used R/meta (v8.2-1) with HKSJ random-effects, as documented in the analysis file. Primary interpretation emphasized consistency across sensitivity assumptions rather than any single ρ value.[19]

3. Results

3.1. Study selection and characteristics

Three contemporary prospective/prospective-like evaluations met inclusion criteria: the randomized MASAI trial (n = 80,020 modified-ITT), the prospective paired-reader ScreenTrustCAD study (n = 55,581), and the nationwide real-world PRAIM implementation (n = 461,818). In aggregate, 597,419 screening examinations contributed to the evidence synthesis[20–23] (Table 1).

When multiple publications described the same trial or overlapping cohorts, they were linked under a single study identifier and were not

Table 1
Study characteristics and program-level outcomes.

Panel A. Study design and workflow.

Study	Design	AI system	Comparison	AI workflow mechanism	N; design note
MASAI (Sweden)	Randomized screening-accuracy trial; modified intention-to-treat safety analysis	Transpara v1.7	AI-supported risk triage vs standard double reading	Risk score 1–10; single/double reading determined by risk; CAD display for highest-risk examinations	80,020; interim safety analysis; interval-cancer endpoint pending
ScreenTrustCAD (Sweden)	Prospective, population-based, paired-reader non-inferiority study	Lunit Insight MMG	One radiologist + AI vs two-radiologist double reading	AI as independent initial reader; positive initial reads proceeded to consensus with AI information available	55,581; screen-positive design; relative CDR/recall comparisons informative; absolute sensitivity/NPV require follow-up
PRAIM (Germany)	Nationwide real-world implementation; overlap-weighted analysis	Vara MG	AI-supported double reading vs standard double reading	Decision-referral workflow with normal triage and safety-net re-review of AI-flagged cases	461,818; observational implementation with overlap weighting to mitigate reader self-selection

Panel B. Primary quantitative outcomes.

Study	CDR AI vs control (/1000)	CDR RD (/1000)	Recall AI vs control (/1000)	Recall RD (/1000)
MASAI	6.1 vs 5.1	+1.0	21.5 vs 20.4	+1.1
ScreenTrustCAD	4.70 vs 4.50	+0.20	28.0 vs 29.3	−1.31
PRAIM	6.7 vs 5.7	+1.0	37.4 vs 38.3	−0.9

Panel C. Secondary workflow and tumor-spectrum signals.

Study	PPV: recall; biopsy	Efficiency/time signal	Tumor spectrum or safety-net signal
MASAI	Recall: 28.3% vs 24.8%; biopsy: not reported	44.3% fewer total readings	In situ cancers more frequent with AI: 25% vs 19% of screen-detected cancers
ScreenTrustCAD	Recall: 16.8% vs 15.3%; biopsy: 64.8% vs 64.8%	AI + one strategy saves one initial radiologist read per examination; recall after consensus 4% lower than standard double reading	AI-only and triple-reading were secondary workflow analyses and were not pooled in the primary synthesis
PRAIM	Recall: 17.9% vs 14.9%; biopsy: 64.5% vs 59.2%	Shorter reads for AI-normal examinations; automation scenario projected 56.7% fewer initial reads	Safety-net accepted in 1077 cases, leading to 541 recalls and 204 recovered cancers

Abbreviations:
AI = artificial intelligence;
CAD = computer-aided detection;
CDR = cancer detection rate;
NPV = negative predictive value;
PPV = positive predictive value;
RD = risk difference.
Values are AI-supported workflow minus control. ScreenTrustCAD PPV values were derived from published counts for the AI + one radiologist versus two-radiologist comparison.

treated as independent studies. Only one report contributed quantitative data for each primary outcome. For MASAI, the 2023 prespecified clinical-safety analysis provided the cancer-detection and recall data used in the quantitative synthesis, whereas the 2025 expanded-cohort publication was treated as a companion report and summarized narratively.

The included studies were:

MASAI randomized screening-accuracy trial (Sweden) evaluating risk triage (AI examination risk score 1–10) combined with limited CAD display (scores 8–10), with single vs double reading determined by risk, and prespecified clinical safety analysis after ~80,000 screens.[19] Key design details: population invitations, PACS-level 1:1 randomization after image acquisition, participants/radiographers masked to allocation, unblinded double reading per standard of care, and modified intention-to-treat primary analysis; primary endpoint for the full trial is interval cancer rate after 2-year follow-up. Among 39,996 participants in the AI-supported arm and 40,024 in standard double reading, screen-detected cancers were 244 vs 203 (6.1 vs 5.1 per 1000; RD +1.0/1000), and recalls were 861 vs 817 (21.5 vs 20.4 per 1000; RD +1.1/1000). False-positive rate was identical (1.5% in both arms). PPV of recall was higher with AI (28.3% vs 24.8%). Workload fell from 83,231 to 46,345 total reads (−44.3%). In situ cancers were more frequent with AI (25% vs 19%).[21]

ScreenTrustCAD was a prospective, population-based, paired-reader, non-inferiority study in Sweden that evaluated AI as an independent reader within a double-reading screening workflow. For the primary contrast used in this synthesis, one radiologist plus AI was compared with standard double reading by two radiologists. Among 55,581 included examinations, AI plus one radiologist detected 261 cancers versus 250 with two-radiologist double reading, corresponding to a CDR RD of +0.20 per 1000 examinations. Recall after consensus occurred in 1556 examinations with AI plus one radiologist versus 1629 with two radiologists, corresponding to a recall RD of −1.31 per 1000 examinations. AI-only and triple-reading strategies were considered secondary workflow analyses and were summarized narratively rather than pooled in the primary meta-analysis.[20]

PRAIM nationwide, multicenter real-world implementation study (Germany) comparing AI-supported double reading with standard double reading using a decision-referral approach: normal triaging (worklist tagging) and a safety-net alert prompting re-review of AI-flagged suspicious regions when the initial reader called a case normal.[23] To address reader self-selection and workflow confounding, primary analyses used propensity score overlap weighting with a pre-specified causal DAG, plus extensive sensitivity analyses (alternative weighting, trimming, bootstrapping, placebo intervention). Reading times were captured in the AI group.[21]

After overlap weighting to address reader self-selection and AI-prediction–linked workflow choices, breast cancer detection rate (BCDR) was 6.7 vs 5.7 per 1000 (RD +1.0/1000; +17.6%, 95% CI +5.7%, +30.8%), and recall was 37.4 vs 38.3 per 1000 (RD −0.9/1000; −2.5%, 95% CI −6.5%, +1.7%). PPV of recall (17.9% vs 14.9%) and PPV of biopsy (64.5% vs 59.2%) were higher with AI (Table 3, p. 921). Reading time in the AI group was shorter for normal-tagged exams (median 16 s; mean 39 s) than for unclassified exams (median 30 s; mean 67 s). The safety-net alert was accepted in 1077 cases (0.4% of all AI-group exams), leading to 541 recalls and 204 cancers that would otherwise have been missed.

3.2. *Quantitative synthesis of primary outcomes*

- Cancer detection rate (CDR) (Table 4)

Pooling per-study risk differences (RDs) per 1000 with HKSJ random-effects and Kish *N*eff adjustment for ScreenTrustCAD's paired design yielded a pooled RD ≈ +0.9 cancers per 1000 screens (Fig. 1). At $\rho = 0.5$ (primary), the HKSJ 95% CI spanned −0.0 to +1.8 per 1000 ($I^2 \approx 12\%$). At higher within-pair correlations ($\rho \geq 0.7$), the 95% CI excluded 0(+0.1 to +1.7 per 1000; $I^2 = 0\%$) [Fig. 1]

- Recall rate (Fig. 2) (Table 4)

The pooled recall RD favored AI but was not statistically definitive: ≈ −0.6 per 1000 (HKSJ 95% CI −3.1 to +2.1 per 1000; $I^2 \approx 41$–43% across $\rho = 0.3$–0.8).[23,24]

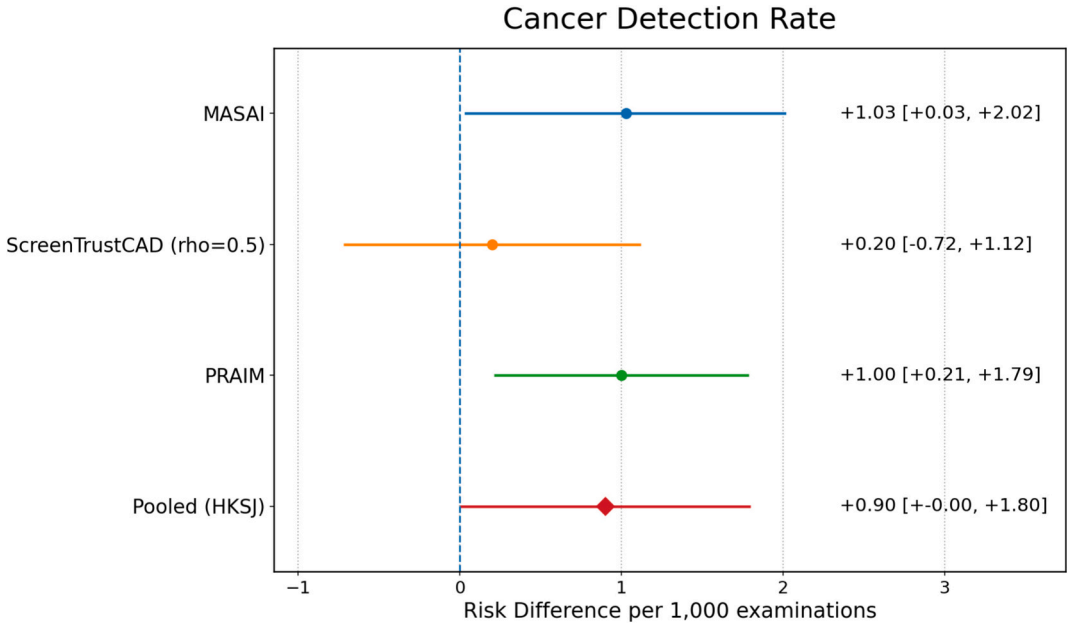


Fig. 1. Cancer detection rate (RD per 1000). Study-specific risk differences (AI - control) and HKSJ random-effects pooled estimate (primary rho = 0.5 Kish effective sample size for ScreenTrustCAD). Horizontal lines indicate 95% CIs.

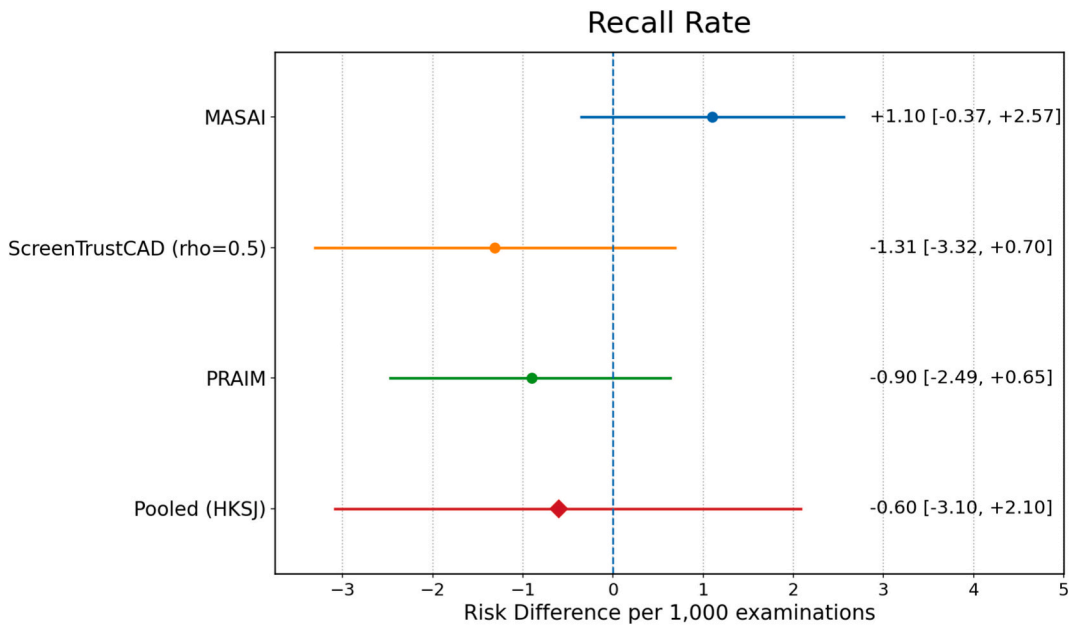


Fig. 2. Recall rate (RD per 1000). Study-specific risk differences (AI - control) and HKSJ random-effects pooled estimate (primary rho = 0.5). Negative values favor AI for recall. Horizontal lines indicate 95% CIs.

3.3. Sensitivity analyses

- ScreenTrustCAD ρ-sensitivity. Varying the assumed within-examination correlation from ρ = 0.3 to 0.8 affected precision without materially changing the study-level point estimates or their direction (CDR RD +0.20/1000; recall RD −1.31/1000).
- Leave-one-out. Excluding ScreenTrustCAD (MASAI+PRAIM) preserved a CDR gain of +1.0/1000 (95% CI +0.9 to +1.1) with neutral recall; excluding PRAIM (MASAI+ScreenTrustCAD) yielded smaller, imprecise CDR gains (+0.6/1000; 95% CI −0.1 to +1.3) with neutral recall.[23]

3.4. Secondary outcomes

- Positive predictive value (PPV). Both MASAI and PRAIM showed higher PPV with AI (MASAI PPV of recall 28.3% vs 24.8%; PRAIM PPV of recall 17.9% vs 14.9%; PPV of biopsy 64.5% vs 59.2%),
- Tumor spectrum. MASAI observed more DCIS with AI (25% vs 19% of screen-detected cancers) with similar invasive detection; PRAIM's model-based stage-stratified estimates showed AI-associated increases particularly at stage 0–1 and grade 1, with no increase in stage 3–4
- Workload and time. MASAI documented a 44.3% reduction in total screen readings, while ScreenTrustCAD's AI + one strategy would save one initial read per exam and lower recall by ~4% versus two

Table 2
Risk of bias and applicability assessment.

Study	Patient selection / confounding	Index test and workflow	Reference standard / outcome verification	Flow and timing	Applicability and main rationale
MASAI	Low risk: randomized allocation within population screening; modified intention-to-treat analysis.	Low risk: AI triage workflow and CAD display were prespecified.	Unclear risk: screen-detected cancer outcomes reported, but interval-cancer endpoint for the full trial remains pending.	Low risk: concurrent comparator and clear screening outcomes.	Low concern: organized Swedish double-reading screening; strongest design among included studies but currently an interim safety analysis.
ScreenTrustCAD	Low risk: consecutive population-based screening participants, with prespecified exclusions.	Low risk: AI threshold calibrated a priori; AI and radiologists read independently at the initial stage.	High/unclear risk: screen-positive design verifies disease status primarily among positives; absolute sensitivity, specificity, and NPV cannot be directly estimated before interval follow-up.	Low risk: paired design applied to the same examinations; 3-month screen-detected cancer endpoint.	Low-to-moderate concern: directly relevant to double-reading plus consensus workflows, but results are specific to local workflow, mammography equipment, and AI product.
PRAIM	High risk: observational implementation with reader self-selection and workflow confounding; overlap weighting mitigates but cannot eliminate residual bias.	Low risk: decision-referral workflow embedded in routine double reading with defined normal-triage and safety-net functions.	Low risk for screen-detected outcomes; longer-term interval-cancer and stage-distribution monitoring remain important.	Low risk: nationwide program data with model-based estimates and sensitivity analyses.	Low-to-moderate concern: highly relevant to organized German double-reading programs, but implementation-specific and not randomized.

Abbreviations
CAD = computer-aided detection;
NPV = negative predictive value;
QUADAS-2 = Quality Assessment of Diagnostic Accuracy Studies version 2.
Ratings emphasize bias relevant to the outcomes pooled in this review: CDR and recall risk differences.

radiologists (Table 2). PRAIM recorded shorter reading times for AI-normal exams (median 16 s vs 30 s) and, in a fictitious automation scenario (normal-tagged exams not read by radiologists), projected ~56.7% fewer initial reads with lower recall and higher BCDR.

3.5. Risk-of-bias

Risk-of-bias assessments are summarized in Table 2.

4. Discussion

Across three large, contemporary evaluations embedded within population mammography screening (MASAI (randomized AI-supported risk triage)), ScreenTrustCAD (prospective paired-reader evaluation with AI as an independent reader), and PRAIM (nationwide decision-referral implementation within double reading), AI support was associated with a modest absolute increase in cancer detection without a consistent increase in recall, together with efficiency signals relevant to program delivery.

In MASAI's prespecified safety analysis (~80,000 examinations), AI was associated with approximately +1 additional cancer per 1000 with an unchanged false-positive rate, higher PPV of recall, and ~44% fewer total readings, with cancers concentrated in the highest-risk stratum (top 1% AI risk examinations).[20–21] A subsequent protocol-defined analysis of the expanded MASAI cohort provided additional evidence consistent with the 2023 clinical-safety analysis used in the quantitative synthesis. In the modified intention-to-treat population of 105,915 participants, AI-supported screening detected 338 cancers compared with 262 with standard double reading, corresponding to cancer-detection rates of 6.4 versus 5.0 per 1000 examinations and a 29% relative increase. Recall rates were 2.1% versus 1.9% and did not differ significantly, and false-positive rates were not increased. Positive predictive value of recall was higher with AI-supported screening (30.5% vs 25.5%), while screen-reading workload was reduced by 44.2%. The additional detected cancers were predominantly small, lymph-node-negative invasive cancers, together with increased detection of in situ disease. These findings strengthen the workflow-level interpretation of MASAI. Because the 2023 and 2025 reports include overlapping participants from the same randomized trial, the 2025 report was not entered as an additional independent study in the meta-analysis.[22–23]

This synthesis is not intended to establish long-term outcomes (e.g., mortality effects or overdiagnosis) from three heterogeneous implementations. Rather, it quantifies near-term program-level tradeoffs (detection, recall, PPV, and workflow) using the best available prospective or prospectively embedded evidence. The incremental value of this synthesis lies in translating heterogeneous prospective screening studies into a common absolute scale relevant to screening-program operations. Although the included AI systems and workflows differ, all three studies address the same practical policy question faced by organized screening programs: whether AI integration changes cancer detection, recall, PPV, and radiologist workload when embedded into population screening. Therefore, the pooled estimates are best understood as order-of-magnitude benchmarks for program planning, while the study-level estimates remain essential for interpreting workflow-specific effects. The leave-one-out analyses suggest that the pooled CDR gain is driven primarily by MASAI and PRAIM, whereas exclusion of PRAIM attenuates the pooled effect. This pattern may reflect differences in how AI outputs are operationalized. MASAI and PRAIM used AI to redirect human attention through risk triage or decision referral, whereas ScreenTrustCAD evaluated AI as an independent reader within a paired-reader framework. These findings raise the possibility that AI may have the greatest near-term effect on detection when embedded into workflows that actively prioritize high-risk examinations or trigger targeted re-review. However, because each implementation model is represented by a single study, this interpretation should be considered hypothesis-generating rather than definitive.

Table 4
Pooled cancer detection and recall rate differences with heterogeneity estimates.

Outcome	Pooled RD (HKSJ)	95% CI	Heterogeneity
Cancer detection rate (CDR)	+0.9 per 1000	−0.0 to +1.8 per 1000	I2 = 12%
Recall rate	−0.6 per 1000	−3.1 to +2.1 per 1000	I2 = 41–43%

Table 3
Sensitivity and robustness analyses.

Analysis	Question addressed	Result to report	Interpretation
Primary model	HKSJ random-effects model with ScreenTrustCAD adjusted using Kish effective sample size at rho = 0.5.	CDR RD +0.9 per 1000 (95% CI −0.0 to +1.8; I2 approx. 12%); recall RD −0.6 per 1000 (95% CI −3.1 to +2.1; I2 approx. 41–43%).	Exploratory pooled estimate suggests a small directional CDR increase without a consistent recall penalty.
ScreenTrustCAD rho-sensitivity	Does the assumed within-examination correlation affect conclusions?	Point estimates unchanged: ScreenTrustCAD CDR RD +0.20 per 1000 and recall RD −1.31 per 1000; rho from 0.3 to 0.8 affected precision but not direction.	Paired-design assumptions affect uncertainty more than the qualitative conclusion.
ScreenTrustCAD nominal-N analysis, no Kish adjustment	Does giving ScreenTrustCAD its full nominal denominator change the pooled result?	CDR estimate attenuated to approximately +0.7 per 1000 because ScreenTrustCAD had the smallest CDR RD; recall remained neutral or favored AI.	Directly addresses reviewer concern that the Kish adjustment dampens ScreenTrustCAD weight.
Leave-one-out: exclude MASAI	Is the pooled result dependent on the randomized MASAI trial?	CDR remained directionally positive but less precise; recall remained neutral or favored AI.	Shows that the pooled estimate is not wholly dependent on MASAI; interpretation remains workflow-specific.
Leave-one-out: exclude ScreenTrustCAD	What happens when the paired-reader independent-reader study is removed?	CDR RD approximately +1.0 per 1000 (95% CI +0.9 to +1.1); recall neutral.	The CDR gain is preserved when pooling MASAI and PRAIM, consistent with stronger effects in risk-triage and decision-referral workflows.
Leave-one-out: exclude PRAIM	What happens when the largest observational implementation is removed?	CDR RD approximately +0.6 per 1000 (95% CI −0.1 to +1.3); recall neutral.	PRAIM contributes materially to the pooled CDR gain; should be discussed as workflow-dependent and hypothesis-generating.
Common-effect model for context	Are results directionally similar under a less conservative model?	Directionally similar pooled RDs with smaller standard errors.	Report only as context; random-effects HKSJ remains primary because designs and workflows differ.

Abbreviations:
CDR = cancer detection rate;
CI = confidence interval;
HKSJ = Hartung-Knapp-Sidik-Jonkman;
RD = risk difference.
Sensitivity analyses are robustness checks and exploratory workflow benchmarks rather than definitive subgroup inference.

ScreenTrustCAD illustrates why AI effects should be interpreted at the workflow level rather than as fixed algorithm-level performance. In that study, AI plus one radiologist increased abnormal interpretations at the initial-read stage, suggesting that AI and human readers flagged partly different examinations. However, this did not translate into higher recall because consensus review filtered AI-positive findings before recall. This supports the concept that the safety and efficiency of AI in screening depend not only on algorithmic sensitivity but also on downstream arbitration, consensus, and safety-net design.

Most prior evidence syntheses on AI in breast cancer screening have focused on standalone diagnostic performance, pooling AUC, sensitivity, and specificity for AI acting as an independent reader, frequently based on retrospective, case–control, or enriched datasets.[24,25] A representative example is the meta-analysis by Yoon et al. (2023), which evaluated standalone AI performance for digital mammography and digital breast tomosynthesis but did not address program-level outcomes such as absolute changes in cancer detection or recall per 1000 women screened.[25] Broader reviews have summarized "AI in screening" across heterogeneous designs and workflows—including standalone AI, second-reader replacement, triage, and decision-support models—but often do not restrict inclusion to prospective population screening programs and therefore cannot pool absolute risk differences consistently.[27–28] As a result, prospective program-level evidence has frequently been synthesized narratively, with studies such as MASAI and PRAIM discussed descriptively rather than quantitatively. Against this background, the present study addresses a distinct gap by restricting inclusion to prospective or prospectively embedded population screening programs and harmonizing outcomes as absolute risk differences per 1000 examinations, with relevance to European screening workflows characterized by double reading and arbitration.

The findings of this synthesis should be interpreted primarily in the context of human–AI pairing strategies—AI as an independent second reader, or AI-based triage approaches that selectively reduce the need for a second human read—rather than as evidence for standalone AI paradigms. These configurations align with contemporary European screening practice and differ fundamentally from retrospective accuracy studies that evaluate AI outside routine workflow constraints. Observed effects on cancer detection, recall, and workload are inherently workflow-dependent and may vary according to arbitration rules, reading thresholds, and safety-net mechanisms, emphasizing that AI performance cannot be meaningfully interpreted outside the human–AI system in which it is embedded.

By expressing effects as absolute risk differences per 1000 examinations, this synthesis aligns reporting with how population screening programs are planned, monitored, and benchmarked. Absolute measures directly inform decisions regarding recall capacity, downstream diagnostic demand, and workforce allocation, whereas relative or accuracy-based metrics alone can obscure the practical implications of AI integration at scale. This framing is particularly relevant for European programs, where even small absolute changes may translate into substantial system-level effects.

The tumor-spectrum findings also warrant attention. MASAI reported a higher proportion of in situ cancers among AI-detected screen cancers, and PRAIM showed AI-associated increases particularly for stage 0–1 and grade 1 disease. Clinically, this suggests that AI-supported workflows may help radiologists identify subtle or early-stage abnormalities that are otherwise more likely to be missed or deferred. However, enrichment for DCIS and low-stage cancers should not be interpreted automatically as net benefit. Longer-term follow-up is needed to determine whether these additional detections translate into lower interval cancer rates, more favorable stage distribution at subsequent screening, or potential overdiagnosis. Future AI-screening studies should therefore report interval cancers, tumor biology, stage, grade, and treatment-relevant outcomes in standardized fashion.

Between-program variability should be interpreted as an expected consequence of workflow heterogeneity, rather than inconsistency of AI performance per se. The included programs differed in AI integration strategy (independent reader, triage, or decision referral), arbitration pathways, and safety-net designs, each of which modulates how AI outputs translate into recalls and diagnoses. Consequently, effects on detection and recall are best understood as properties of the human–AI workflow, rather than fixed attributes of the algorithm alone.

Safety is central to AI adoption in screening. Although long-term outcomes such as interval cancer rates and mortality cannot be established from the present synthesis, available prospective evidence provides reassurance that AI-supported workflows can be implemented without a consistent recall increase and with explicit safety-net mechanisms. In PRAIM, for example, decision-referral was paired with safeguards that recovered cancers that would otherwise have been missed, underscoring the importance of embedding AI within fail-safe program designs and pairing implementation with explicit quality assurance and ongoing monitoring of interval cancers and stage distribution.

From a radiologist perspective, the principal value of AI in double-reading screening programs may lie less in replacing human interpretation than in redistributing cognitive effort. By concentrating human attention on higher-risk examinations and reducing redundant reads in low-risk cases, AI-supported workflows may preserve diagnostic vigilance while alleviating workload pressures. This reframing may be particularly relevant in settings facing workforce shortages, increasing screening volumes, or delays in reporting and arbitration.

Future evaluations should prioritize standardized reporting of interval cancer rates, stage distribution, and arbitration outcomes, alongside transparent descriptions of workflow configuration. Comparative studies of different human–AI pairing strategies within the same screening infrastructure may clarify which models optimize safety, efficiency, and acceptability across diverse European programs.

4.1. Limitations

Several limitations should be considered. First, the number of eligible prospective or prospectively embedded studies remains small, limiting the strength and precision of pooled inference. Second, the included studies differed in design, AI product, population, and implementation strategy; therefore, pooled estimates should be interpreted as exploratory program-level benchmarks rather than as a universal AI effect. Third, formal pooling by AI use-case was not statistically meaningful because each major implementation strategy was represented by a single study. Fourth, ScreenTrustCAD used a screen-positive design, which precludes direct estimation of absolute sensitivity, specificity, or negative predictive value before interval-cancer follow-up, although relative cancer detection and downstream recall comparisons remain informative for workflow evaluation. Fifth, PRAIM used model-based overlap weighting to address reader self-selection and workflow confounding, but residual bias cannot be fully excluded. Sixth, the MASAI cancer-detection and recall data used in the primary quantitative synthesis were drawn from the 2023 clinical-safety analysis, whereas the later overlapping expanded-cohort report was incorporated narratively rather than substituted into the pooled analysis. Finally, the meta-analysis used aggregate published data and assumptions regarding the paired-reader correlation structure in ScreenTrustCAD; this was addressed through ρ-sensitivity and nominal-N sensitivity analyses.

5. Conclusion

In contemporary European-style population mammography screening programs, prospective and program-embedded evidence suggests that AI-supported workflows may yield a small absolute increase in cancer detection without a consistent increase in recall, alongside improved PPV and workflow-efficiency signals. Because the current evidence base consists of only three heterogeneous implementations, these pooled estimates should be interpreted as exploratory program-level benchmarks rather than as a universal AI effect. The

findings support continued evaluation of AI as a complementary component of double-reading workflows, with implementation tied to explicit safety-net mechanisms, quality assurance, interval cancer monitoring, and standardized reporting of tumor stage and biology.

CRediT authorship contribution statement

Romuald Ferre: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Thad Benefield:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Cherie M. Kuzmiak:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Funding

None.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The study was conducted independently, and no external funding sources or commercial entities had any role in the study design, data collection, analysis, interpretation, or manuscript preparation.

AI-assisted language tools were used solely for manuscript editing, formatting, and clarity. All study design, data extraction, statistical analyses, and interpretation were performed independently by the authors, who take full responsibility for the integrity and accuracy of the work.

Data availability

All data used in this meta-analysis were extracted from previously published studies. No new patient-level data were generated.

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